

Perplexity and Fluency after Continual Chinese Adaptation of a 42M-Parameter Llama-2

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Abstract

I evaluate English generation, cross-lingual perplexity, and Chinese continual pre-training with a 41.69M-parameter Llama-2-style model trained on TinyStories. The generation study contains 72 continuations from four prompt types and six decoding settings. Beam search is stable but repeats the same output across seeds. High-temperature nucleus sampling produces more varied text, with weaker causal links in several samples. On the official test sets, the base model obtains token-weighted perplexity (PPL) 3.14 in English and 117,885 in Chinese. After 1,200 Chinese updates, Chinese PPL falls to 4.80 and English PPL rises to 7.07. Adding 1,000 English rehearsal stories to the 10,000 Chinese stories gives Chinese PPL 5.05 and English PPL 2.86. The generated Chinese improves at the phrase level, although longer stories still contain changes in character identity and implausible events.

1 Introduction

TinyStories shows that very small language models can learn fluent, multi-paragraph English when their training domain and vocabulary are tightly controlled (Eldan and Li, 2023). The same specialization creates an informative continual learning problem: can an English model acquire Chinese without destroying what it already knows? I investigate this with a 42M Llama-2-style causal model (Touvron et al., 2023; Karpathy, 2023).

I compare six decoding settings across four prompt types, then evaluate the same checkpoint on English, Chinese, and Portuguese with matched token budgets. Continual training holds the update count and optimizer settings fixed while varying the learning rate and corpus composition. The final run adds a small English replay sample to test whether it can preserve the original language while the model learns Chinese.

2 Experimental Setup

Data

I use the supplied local checkpoint and official archive: 10,000/500/1,000 Chinese train/development/test stories, 1,000 English test stories, and matched Portuguese splits. The course archive contains no English training split, so the replay condition uses 1,000 separately sampled TinyStories training stories (Gabrui, 2024). Seed 7021, source labels, and SHA-256 hashes are recorded. All reported evaluation and training results use the supplied checkpoint and official test files.

Model and Tokenization

The supplied checkpoint contains 41,689,600 parameters and uses the original 32K SentencePiece tokenizer. No vocabulary items are added. Chinese requires 1.595 tokens per visible character, compared with 0.267 for English, a $5.95\times$ fragmentation penalty. Thus the same 256-token block covers much less visible Chinese context. This agrees with prior findings that tokenizer quality can materially change multilingual model performance (Rust et al., 2021).

Stories are separated by EOS and packed into 256-token blocks. Evaluation uses the same 128 blocks (32,640 predicted tokens) for every checkpoint. I accumulate token-weighted negative log-likelihood L and report $\exp(L)$. Packing eliminates padding as a confound; using fixed evaluation blocks makes all model comparisons paired.

Split	Docs	Median chars	Mean tokens
Chinese train	10,000	239	405.1
Chinese dev	500	240	401.0
Chinese test	1,000	242	408.8
English test	1,000	738	206.1
Portuguese test	1,000	703	256.3

Table 1: Corpus statistics under the unchanged base tokenizer.

Decoder	Dist-2 $\uparrow$	Rep-4 $\downarrow$	Unique open.
Greedy	.252	.0176	.333
Beam-4	.270	.0083	.333
Low temp. (.45)	.678	.0210	1.000
Top- k 50	.725	.0091	1.000
Nucleus $p = .90$	.712	.0082	1.000
High temp. ($p = .95$)	.874	.0008	1.000

Table 2: Decoding statistics from 12 outputs per setting.

3 English Generation

The prompt set contains (i) a conventional story opening, (ii) dialogue, (iii) an explicit “no magic” constraint, and (iv) a counterfactual village. Each is decoded three times per setting. Greedy and four-beam search serve as deterministic controls. Sampling settings vary temperature, top- k , and nucleus probability. I report corpus-level Distinct-2, within-text repeated four-gram fraction, and the fraction of unique first sentences.

Beam search produced a clean key-and-door narrative, but its three runs for each prompt were identical. High-temperature nucleus sampling achieved the best lexical diversity and lowest exact repetition. Case inspection also found abrupt entity changes and weaker causal links. Top- k 50 gave the most useful balance in this experiment, with high diversity and fewer serious coherence failures. None of the 72 samples emitted EOS within 128 new tokens. The fixed cutoff determines their similar lengths and prevents this run from measuring learned stopping behavior.

Prompt form also mattered. Dialogue openings were usually continued as dialogue, while the instruction-like constraint was often treated as part of the story. Its behavior follows the TinyStories continuation distribution, so instruction-like prompts do not reliably impose negative constraints on this base model.

4 Perplexity and Continual Training

The base model’s fixed-budget English PPL is 3.14. Chinese test PPL is 117,885.17 (development: 111,207.01), over four orders of magnitude higher; Portuguese is also poor at 26,522.53. A Chinese prompt immediately elicits English before adaptation. The scale of this gap and the tokenizer fragmentation show that language mismatch is the main problem.

I tune learning rate with identical 80-update runs. All runs use AdamW, batch size 8, two-step gradient accumulation (4,096 tokens/update), 6% linear warm-up, cosine decay, weight decay 0.1, gradient clipping at 1.0, and bfloat16. Table 3 selects 2×10^{-4} by Chinese development PPL; the larger rate converges quickly but is slightly worse under the same budget.

Two final runs use 1,200 updates. The Chinese-only corpus is compared with a shuffled mix of 10,000 Chinese and 1,000 English rehearsal stories. Rehearsal follows the general continual learning idea of preserving representative old-task examples (Lopez-Paz and Ranzato, 2017); it adds no model parameters and both conditions receive the same optimizer-step budget.

Chinese-only training reduces fixed-budget target PPL by $24,555\times$, while English becomes $2.25\times$ worse.

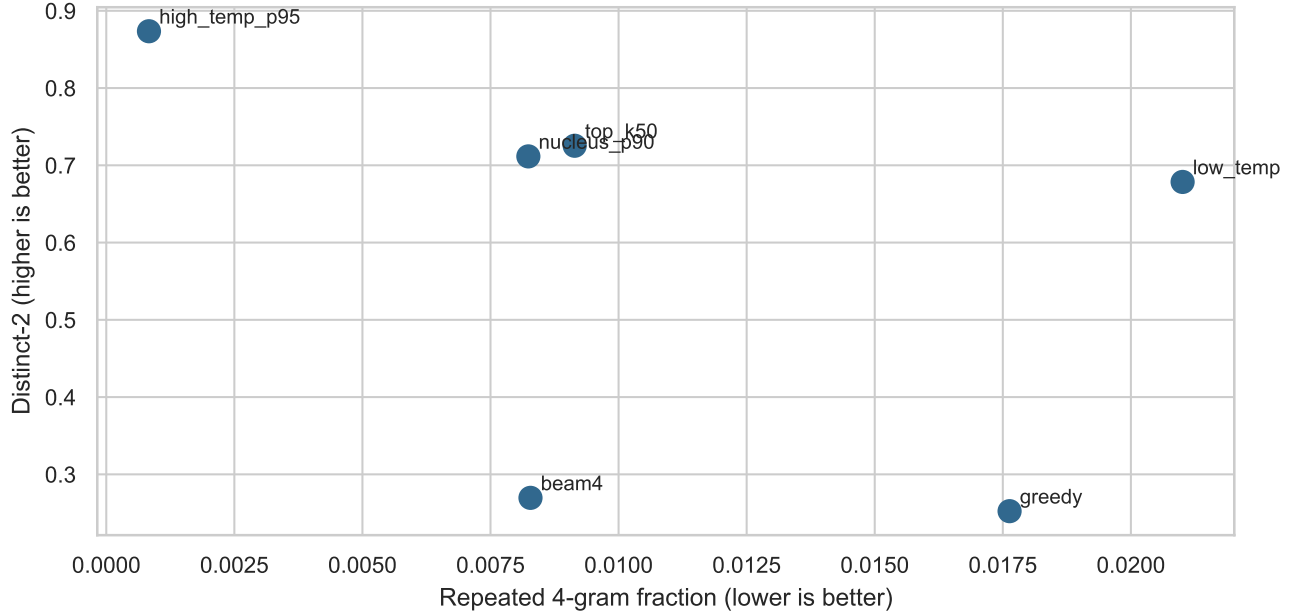


Figure 1: Diversity and repetition across all 12 outputs for each decoding setting.

Learning rate	Dev PPL	Time (s)
5×10^{-5}	412.55	12.4
2×10^{-4}	151.27	12.8
8×10^{-4}	154.39	12.5

Table 3: Equal-budget learning-rate sweep (80 updates each).

Portuguese PPL also rises to 214,567, which shows that the update causes language-specific interference. With rehearsal, Chinese PPL is 5.1% higher than the Chinese-only result and English PPL is 59.6% lower. Fixed-token English PPL is below the base result, while document-mean English PPL remains above it. The document mean gives every story equal weight and is more affected by short, difficult examples. Fixed packing gives every predicted token equal weight. Both metrics show that rehearsal preserves English better than Chinese-only training.

5 Perplexity and Generated Story Quality

After adaptation, 72–90% of visible generated characters are Chinese. The outputs also acquire Chinese punctuation, dialogue, names, and paragraph structure, and some short clauses read naturally. Across longer passages, characters change identity, objects receive incompatible actions, and several endings stop mid-byte at the fixed cutoff. The rehearsal run shows the same pattern.

Teacher-forced likelihood rewards likely local continuations even when errors accumulate during free generation. The byte-heavy tokenizer can also make familiar short fragments easy to predict while reducing the amount of visible context in each block. Model size and the single packed training epoch impose further limits. In this experiment, PPL tracks local adaptation more reliably than overall story quality, so the generations need a separate qualitative or human evaluation.

Checkpoint	Fixed-token		Document mean	
	ZH	EN	ZH	EN
Base	117,885.17	3.14	76,898.29	3.53
Chinese-only	4.80	7.07	4.86	8.84
ZH+EN rehearsal	5.05	2.86	5.77	4.38

Table 4: Official test PPL. Fixed-token uses 32,640 predictions; document mean follows `compute_ppl` with a 512-token cap.

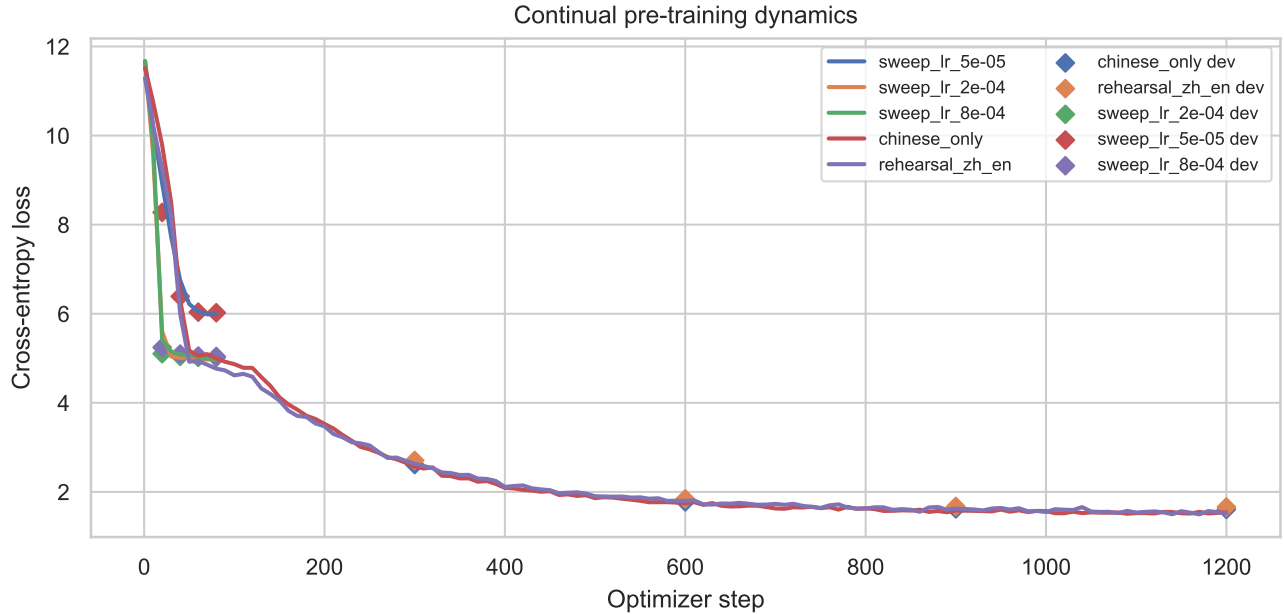


Figure 2: Training loss (lines) and fixed Chinese development loss (diamonds). Curves include the sweep and both final runs; logs are recorded every 10 updates.

6 Conclusion

The decoding comparison shows a clear trade-off between varied wording and stable story structure. Top- k 50 gives the best balance among the tested settings. Continual training closes most of the Chinese likelihood gap, but Chinese-only training damages English and Portuguese performance. A 9.1% story-level English rehearsal share keeps English close to its original level for a 5.1% increase in Chinese PPL. This makes the rehearsal checkpoint the better choice when both Chinese adaptation and English retention matter. Future work should test a Chinese-aware tokenizer, collect human ratings from multiple assessors, and use longer generations to measure EOS behavior.

References

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